ROOF · RTLS *what* **PERIOD 3** *how* SKATE · IMU

WHICH STRIDE BROKE DOWN - AND WHEN?

Fatigue vs bad biomechanics in ice hockey · Hugo Persson & Thrainn Leo Thorisson · MSc Sports Technology, KTH

01 THE PROBLEM

50% EUROPEAN

20% NCAA

- **Groin and hip injuries are the biggest problem in ice hockey**, reported by half of European professional players and one in five NCAA players. Strength imbalance and reduced hip range of motion are known risk factors [1].
- In the NHL, players whose hip adductor strength was below 80% of their abductor strength were **17x more likely** to suffer a groin strain [2].
- Fatigue also changes how players skate. As players tire, their joint angles fluctuate more, their legs move less in sync and their trunk position shifts. **Wearable sensors can detect these changes** [3].
- We know poor mechanics raise injury risk, and that fatigue causes poor mechanics. **No study has yet linked the two on real ice. That is the gap our system targets.**

02 OUR SYSTEM: TWO LAYERS, ONE STRIDE

REAL-TIME LOCATING SYSTEM · RTLS

Tags in the players' shoulder pads are tracked by **receivers mounted in the arena roof**. Every shift is split into a sequence of **skating actions [4]**:

- **Forward- & Backward strides**
- **Crossovers & Turns**
- **Glides & Stops**

This tells us what the player is doing during a game.

SKATE IMUS · HOW

An IMU on each skate detects when the blade hits and leaves the ice, giving raw data for **every stride** [5].

- **Contact time and stride time** – from the skate IMUs.
- **Stride length** – most accurate with an extra IMU on the pelvis [5].
- **Left/right asymmetry** – by comparing the two skates.

This tells us how each stride is performed.

Forward · Backward · X-over R · **X-over L** · Glide · C-cuts · Left turn · Right turn · Stop

Fused: every stride gets a label. A bad stride is flagged with which action, which shift - and whether it looks like fatigue or technique.
Actions after Swarén et al. [4].
Positioning and IMUs have already been combined on real ice, but to recognise shot types, not to monitor skating [6].

Together, the two layers show not just that a player is tiring, but which movement is breaking down.

03 COACH SEES

1. The **RTLS** splits every shift into skating actions, showing what each player does on the ice.
2. The skate **IMUs** measure each stride within those actions and compare it to the player's own baseline.
3. The **app** gathers the data. When a player's strides drift outside their baseline they are flagged, with the skating action that is breaking down and whether it looks like fatigue or poor technique.



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